

Information Gain and Neural Growth: Classical Identities and Explicit Boundaries

Phase 7 research note

October 7, 2026

This note is a supporting mathematical toolbox. It makes no claim that the identities, divergence bounds, or elementary examples constitute a new autonomous-growth theorem.

1 What information gained by a representation does and does not buy

All logarithms are natural. Representations and the target are random variables on standard Borel spaces; conditional laws exist. The target is finite except where real regression is stated explicitly. Population risks are evaluated on a fresh example. For learned encoders, all statements can be applied conditional on a fixed training history, provided the fresh example has the stipulated conditional law. Conditioning does not assert that the encoder is independent of its training examples.

Write $B_\ell(Z) = \inf_g \mathbb{E}\ell(Y, g(Z))$, where the infimum runs over all measurable decoders with the appropriate action space. A deterministic refinement has $Z = \pi(Z')$ almost surely. More generally, the conclusions stated for refinements hold when $Y - Z' - Z$ is a Markov chain under the specified coupling. Appending a feature is a refinement; retraining and replacing an entire hidden layer need not be one. The results in this section are classical identities, elementary consequences, and explicit counterexamples. Proper-loss information and Bregman interpretations are established in [4, Theorems 8 and 10] and [2, Theorem 1 and Section 2.2].

Proposition 1 (Information gain includes a possible forgetting term). *For finite Y , logarithmic loss gives $B_{\log}(Z) = H(Y | Z)$. For any pair of representations on a common probability space,*

$$B_{\log}(Z) - B_{\log}(Z') = I(Y; Z' | Z) - I(Y; Z | Z').$$

For a refinement, the second term vanishes, and the Bayes improvement is exactly $I(Y; Z' | Z)$.

Proof. Conditional expected log loss at $Z = z$ equals the entropy of the true conditional distribution plus its KL divergence to the prediction. The divergence is nonnegative and is zero at the true distribution. Next expand both conditional mutual informations, using $H(Y | Z, Z')$ as their common subtracted term. Finally the Markov condition makes $I(Y; Z | Z') = 0$. $\square$

Thus positive conditional mutual information concerns an optimal unrestricted decoder; it is not a decrease promised for a trained decoder. Nor does adding and then reoptimizing neurons automatically preserve the information-refinement condition.

Proposition 2 (Squared and classification risk under refinement). *Suppose $Y - Z' - Z$ is Markov. If Y is real with $\mathbb{E}Y^2 < \infty$, put $m = \mathbb{E}[Y | Z]$ and $m' = \mathbb{E}[Y | Z']$. Then*

$$\Delta_{\text{sq}} := B_{\text{sq}}(Z) - B_{\text{sq}}(Z') = \mathbb{E}(m' - m)^2.$$

If additionally $Y \in [a, b]$ almost surely and $J = I(Y; Z' | Z)$, then

$$0 \leq \Delta_{\text{sq}} \leq \frac{1}{2}(b-a)^2 J.$$

For finite-label classification with zero-one loss,

$$0 \leq \Delta_{01} := B_{01}(Z) - B_{01}(Z') \leq \sqrt{J/2}.$$

For $Y \in \{0, 1\}$, both constants are asymptotically sharp among refinements. Here the Brier loss means $(Y - p)^2$, not the sum over both class coordinates. If the binary coarse posterior satisfies $m \in [\alpha, 1 - \alpha]$ almost surely, $0 < \alpha \leq 1/2$, then also

$$J \leq \frac{\Delta_{\text{sq}}}{\alpha(1 - \alpha)}.$$

The last bound is a convenient reverse bound, not a sharp constant claim.

Proof. Conditional expectation minimizes squared loss. The tower identity and the Markov condition give $\mathbb{E}[m' | Z] = m$, hence $\mathbb{E}[m'm] = \mathbb{E}m^2$. Subtracting the two Bayes risks gives $\mathbb{E}(m')^2 - \mathbb{E}m^2 = \mathbb{E}(m' - m)^2$.

Write $P' = P_{Y|Z'}$ and $P = P_{Y|Z}$. Under the Markov condition, $J = \mathbb{E}\text{KL}(P' \| P)$. The mean difference for a variable in $[a, b]$ is at most $(b-a)\text{TV}(P', P)$. Pinsker's inequality, in nats, states $\text{TV}(P', P)^2 \leq \text{KL}(P' \| P)/2$. Integration proves the squared-risk bound. The conditional best classification accuracy is $\max_y P(y)$. Since $\max_y P'(y) - \max_y P(y) \leq \text{TV}(P', P)$, integration, Cauchy-Schwarz, and Pinsker give the upper bound. The nonnegative sign follows from conditional Jensen, since $P = \mathbb{E}[P' | Z]$ and the maximum is convex.

For sharpness let Z be constant and let $m' = 1/2 \pm \epsilon$ with equal probabilities. Then $\Delta_{\text{sq}} = \epsilon^2$, $\Delta_{01} = \epsilon$, and $J = \log 2 - h(1/2 + \epsilon) = 2\epsilon^2 + O(\epsilon^4)$, where $h(p) = -p \log p - (1-p) \log(1-p)$. Finally $\log u \leq u - 1$ gives $\text{kl}(q, p) \leq (q-p)^2/[p(1-p)]$ for $0 < p < 1$. Apply this conditionally and integrate. $\square$

The bounded-response implication is an instance of the classical information-to-conditional-mean inequality; see the earlier absolute-mean version in [6, Lemma 4.4] and the explicit sharp squared-mean formulation in [3, Corollary 7.11]. Pinsker and reverse-divergence comparisons are established tools [5, Eq. (1), Section VI]. These statements are not new neuron-growth guarantees.

Proposition 3 (Function preservation and representation preservation differ). *If $Z' = F(Z)$ and $Z = G(Z')$ almost surely for measurable F, G , then the two representations generate the same sigma field and $I(Y; Z') = I(Y; Z)$, $I(Y; Z' | Z) = 0$. Exact neuron duplication before symmetry-breaking perturbations has this property for the complete hidden representation.*

Nevertheless, a function-preserving neural birth can have strictly positive representation information gain. Fix $0 < \delta < 1/2$, let X be uniform on $\{-1, 1\}$, and set

$$\mathbb{P}(Y = 1 | X = 1) = 1 - \delta, \quad \mathbb{P}(Y = 1 | X = -1) = \delta.$$

Start with constant representation $Z = 1$ and prediction $f = 1/2$. Append the ReLU feature $r(X) = \max\{X, 0\}$ with output coefficient zero. The new representation is $Z' = (1, r(X))$, while the new prediction is still $f' = 1/2 + 0r(X) = 1/2$. Then

$$I(Y; Z' | Z) = \log 2 - h(\delta) > 0, \quad \Delta_{\text{sq}} = (1/2 - \delta)^2.$$

The actual prediction risk at birth is unchanged for every loss for which it is defined. In particular its squared risk is $1/4$, its log risk is $\log 2$, and its classification error under the same fixed tie rule is $1/2$, both before and after birth.

Proof. Mutual measurability gives equality of sigma fields, hence equality of all conditional laws. In the neural example $r(X)$ determines X . The marginal label is balanced, with entropy $\log 2$, and its conditional entropy given X is $h(\delta)$. Its conditional mean is $\delta + (1 - 2\delta)r(X)$, at distance $1/2 - \delta$ from $1/2$. Propositions 1–2 give the asserted Bayes gains. The realized predictions coincide pointwise, which proves equality of their realized risks directly. $\square$

Function-preserving duplication is established in Net2Net [1, Section 2.3]; the distinction here is between the unchanged output function and what an unrestricted decoder could recover from the enlarged hidden vector. In the displayed example even an affine decoder can express the new Bayes predictor, but the birth operation has not fitted its coefficients.

Proposition 4 (Four explicit failures of information-only criteria). *The following claims hold, each under a deterministic refinement.*

1. *Activation entropy can be arbitrarily large while task information and every Bayes-risk improvement are zero. Let U be uniform on $\{1, \dots, M\}$, independent of $Y \sim \text{Ber}(1/2)$. Take $Z = 1$ and $Z' = \text{ReLU}(U) = U$. Then $H(Z') = \log M$, $I(Y; Z' | Z) = 0$, and the conditional target law is unchanged.*
2. *Positive label information need not improve classification. Let $Z = 1$, and let Z' be equiprobable with binary label posteriors $1/10$ and $3/10$. Strict concavity gives $J = h(1/5) - \{h(1/10) + h(3/10)\}/2 > 0$, but both Bayes classification errors are $1/5$, since the Bayes action stays zero. This rules out a universally positive classification-gain lower bound at every positive information level; it does not rule out all bounds at larger information levels.*
3. *Without restricting the response or the decision loss, even a fixed positive information gain need not reduce squared Bayes risk. Let $U \sim \text{Ber}(1/2)$ and independent S be uniform on $\{-1, 1\}$. Set $Y = S(1 + U)$, $Z = 1$, and $Z' = U$. Then $U = |Y| - 1$, so $J = \log 2$, but both conditional means are zero and both squared Bayes risks equal $\mathbb{E}Y^2 = 5/2$.*
4. *Fixed structural information cost does not imply useful risk reduction. For independent equiprobable $S \in \{-1, 1\}$ and uniform $U \in \{1, \dots, M\}$, let $X = (S, U)$, $Z = 1$, $Z' = X$, and $Y | X \sim \text{Ber}(1/2 + \epsilon S)$ with $0 < \epsilon < 1/2$. Then*

$$I(X; Z' | Z) = \log(2M), \quad J = \log 2 - h(1/2 + \epsilon), \quad \Delta_{\text{sq}} = \epsilon^2, \quad \Delta_{01} = \epsilon.$$

All three Bayes gains tend to zero as $\epsilon \downarrow 0$ while the structural information cost is fixed and can be arbitrarily large.

Proof. Independence proves the first item. The second follows by averaging the two conditional errors and entropies. The third follows from independence and the zero mean of S . In the fourth, Z' reveals X , so its input information is $H(X) = \log(2M)$. The label is balanced, the conditional entropy is $h(1/2 + \epsilon)$, the posterior mean is $1/2 + \epsilon S$, and the fine classification error is $1/2 - \epsilon$. These observations give all displayed formulas. $\square$

Even for binary labels there is no distribution-free positive constant c with $\Delta_{\text{sq}} \geq cJ$ for all refinements. Indeed let $X = Y \sim \text{Ber}(p)$, $Z = 1$, $Z' = X$. Then $\Delta_{\text{sq}} = p(1 - p)$ and $J = h(p)$, whose ratio tends to zero as $p \downarrow 0$. This is only a failure of a uniform linear comparison; it is not a claim that no nonlinear comparison exists.

Proposition 5 (The exact gap between Bayes value and fitted risk). *For $t \in \{0, 1\}$, fix representation Z_t , an admissible decoder class $\mathcal{G}_t$, and its fitted member g_t . Assume all risks and infima below are finite. Let $R_t(g) = \mathbb{E}\ell(Y, g(Z_t))$, $B_t = B_\ell(Z_t)$, and define*

$$A_t = \inf_{g \in \mathcal{G}_t} R_t(g) - B_t \geq 0, \quad F_t = R_t(g_t) - \inf_{g \in \mathcal{G}_t} R_t(g) \geq 0.$$

Then

$$R_0(g_0) - R_1(g_1) = (B_0 - B_1) + (A_0 - A_1) + (F_0 - F_1).$$

If $\hat{R}_t$ is an empirical criterion, an exact further split is

$$F_t = O_t + E_t, \quad O_t = \hat{R}_t(g_t) - \inf_{g \in \mathcal{G}_t} \hat{R}_t(g) \geq 0,$$

$$E_t = R_t(g_t) - \hat{R}_t(g_t) + \inf_{g \in \mathcal{G}_t} \hat{R}_t(g) - \inf_{g \in \mathcal{G}_t} R_t(g).$$

Here E_t can have either sign. If $U_t = \sup_{g \in \mathcal{G}_t} |R_t(g) - \hat{R}_t(g)| < \infty$, then $|E_t| \leq 2U_t$.

Proof. Add and subtract each infimum to obtain the identities. Uniform deviation bounds both the fitted function's risk difference and the difference of the two infima by U_t in absolute value. $\square$

A_t is representation-decoder approximation error; F_t is the realized population fitting gap, not a pure optimization error. Only O_t is the empirical optimization residual in the displayed split. The algebra holds for data-dependent encoders and classes after they are fixed, but obtaining a small probabilistic bound for U_t then requires appropriate uniform control or fresh data. The identities do not supply one. A larger realizable function class can lower its infimum while finite optimization fails to lower the fitted risk.

Proposition 6 (A total information budget requires nested retention). *Let $Z_0, \dots, Z_T$ be deterministic successive refinements of X . Then*

$$\sum_{t=1}^T I(Y; Z_t \mid Z_{t-1}) = I(Y; Z_T \mid Z_0) = B_{\log}(Z_0) - B_{\log}(Z_T) \leq I(Y; X \mid Z_0).$$

If all information quantities are finite and $C = \sum_{t=1}^T I(X; Z_t \mid Z_{t-1})$, then also

$$C = I(X; Z_T \mid Z_0), \quad B_{\log}(Z_0) - B_{\log}(Z_T) \leq C.$$

Write $M = \min\{I(Y; X \mid Z_0), C\}$. For bounded real $Y \in [a, b]$, the endpoint squared Bayes improvement is at most $(b - a)^2 M / 2$. The corresponding finite-label classification bound is $\sqrt{M/2}$.

Proof. Each earlier representation is a function of every later one. Apply the mutual-information chain rule and telescope the conditional entropies of the finite target. Apply the same chain rule to X (via mutual information, without requiring finite differential entropies). Conditional data processing gives $I(Y; Z_T \mid Z_0) \leq I(Y; X \mid Z_0)$ and $I(Y; Z_T \mid Z_0) \leq I(X; Z_T \mid Z_0)$. Proposition 2 applied once to the endpoint pair proves the remaining bounds. $\square$

The same chain-rule conclusions hold for an explicitly specified joint Markov degradation chain $Y - X - Z_T - \dots - Z_0$. They need not hold for arbitrary successive fitted representations. For example, take a balanced binary $Y = X$ and alternate $Z_t = X$ at odd t and a constant at even t , starting with constant Z_0 . Every odd transition contributes $\log 2$ to $I(Y; Z_t \mid Z_{t-1})$; every even transition contributes zero. Their sum grows without bound as the number of cycles grows, although the final representation after any full cycle has no label information. The omitted reverse conditional information accounts exactly for each forgetting step. A cumulative transcript restores nesting, but describes the information of that retained transcript, not automatically that of the deployed network. None of these chain-rule facts controls optimizer descent or charges finite-precision storage, feature-search computation, or parameter updates.

For a learned sequence, a valid conditional version fixes one common training history H containing the randomness that specifies every encoder, and uses a fresh evaluation sample independent of H . Apply the preceding propositions under each conditional law given H , then average. Conditioning separately on a growing H_t at each stage without accounting for the newly learned information is not the same argument. In particular the randomness specifying the next encoder must be conditioned on, or represented explicitly as part of the observation, before applying a fixed-encoder Bayes comparison.

References

- [1] Tianqi Chen, Ian Goodfellow, and Jonathon Shlens. Net2Net: Accelerating learning via knowledge transfer. In *International Conference on Learning Representations*, 2016. URL <https://arxiv.org/html/1511.05641v4>.
- [2] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437. URL <https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jasa.pdf>.
- [3] Yury Polyanskiy and Yihong Wu. Information theory, 2024. URL <https://people.lids.mit.edu/yp/homepage/data/itbook-export.pdf>. Author-hosted manuscript, August 16, 2024; Corollary 7.11.
- [4] Mark D. Reid and Robert C. Williamson. Information, divergence and risk for binary experiments. *Journal of Machine Learning Research*, 12:731–817, 2011. URL <https://www.jmlr.org/papers/volume12/reid11a/reid11a.pdf>.
- [5] Igal Sason and Sergio Verdú. f -divergence inequalities. *IEEE Transactions on Information Theory*, 62(11):5973–6006, 2016. URL <https://arxiv.org/html/1508.00335v4>.
- [6] Terence Tao. Szemerédi’s regularity lemma revisited. *Contributions to Discrete Mathematics*, 1(1):8–28, 2006. doi: 10.11575/cdm.v1i1.61900. URL <https://cdm.ucalgary.ca/article/download/61900/46638/176530>.